Civil Engineering

Apply Algebra and Geometry

Assignment 1 - Questions Only

Instructions: Answer all questions. Show formula, substitution, calculation, units and interpretation.

Section A: Formula and Concept Questions

1. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$a^m \times a^n = a^{m+n}$

2. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$a^m / a^n = a^{m-n}$

3. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$\log(ab) = \log a + \log b$

4. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$m = (y_2 - y_1)/(x_2 - x_1)$

5. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

Section B: Applied Questions

Problem Set 1

6. Solve a civil engineering problem involving survey control points. Show formula, substitution, correct units and a technical interpretation. (10 marks)

7. Solve a civil engineering problem involving road alignment coordinates. Show formula, substitution, correct units and a technical interpretation. (10 marks)

8. Solve a civil engineering problem involving structural layout grids. Show formula, substitution, correct units and a technical interpretation. (10 marks)

9. Solve a civil engineering problem involving slope and gradient calculations. Show formula, substitution, correct units and a technical interpretation. (10 marks)

10. Solve a civil engineering problem involving area subdivision. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 2

11. Solve a civil engineering problem involving survey control points. Show formula, substitution, correct units and a technical interpretation. (10 marks)

12. Solve a civil engineering problem involving road alignment coordinates. Show formula, substitution, correct units and a technical interpretation. (10 marks)

13. Solve a civil engineering problem involving structural layout grids. Show formula, substitution, correct units and a technical interpretation. (10 marks)

14. Solve a civil engineering problem involving slope and gradient calculations. Show formula, substitution, correct units and a technical interpretation. (10 marks)

15. Solve a civil engineering problem involving area subdivision. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 3

16. Solve a civil engineering problem involving survey control points. Show formula, substitution, correct units and a technical interpretation. (10 marks)

17. Solve a civil engineering problem involving road alignment coordinates. Show formula, substitution, correct units and a technical interpretation. (10 marks)

18. Solve a civil engineering problem involving structural layout grids. Show formula, substitution, correct units and a technical interpretation. (10 marks)

19. Solve a civil engineering problem involving slope and gradient calculations. Show formula, substitution, correct units and a technical interpretation. (10 marks)

20. Solve a civil engineering problem involving area subdivision. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 4

21. Solve a civil engineering problem involving survey control points. Show formula, substitution, correct units and a technical interpretation. (10 marks)

22. Solve a civil engineering problem involving road alignment coordinates. Show formula, substitution, correct units and a technical interpretation. (10 marks)

23. Solve a civil engineering problem involving structural layout grids. Show formula, substitution, correct units and a technical interpretation. (10 marks)

24. Solve a civil engineering problem involving slope and gradient calculations. Show formula, substitution, correct units and a technical interpretation. (10 marks)

25. Solve a civil engineering problem involving area subdivision. Show formula, substitution, correct units and a technical interpretation. (10 marks)

References and Further Reading

These notes are original learning materials prepared for Mr Mwirigi Mathematics Training Hub. They are academically informed by standard engineering mathematics texts and the local TVET curriculum framework. No textbook pages or worked examples have been copied.

1. Stroud, K. A. and Booth, D. J. Engineering Mathematics, 8th edition. Bloomsbury Academic.

2. Bird, J. Engineering Mathematics. Routledge / Taylor & Francis.

3. Bird, J. Higher Engineering Mathematics. Routledge / Taylor & Francis.

4. Relevant Civil Engineering, Building Technology and Land Survey curriculum and occupational standard documents.